Executive summary

This report brings together two related but distinct analyses from the saved seed-2024 owner run: (1) the multistage market-offering optimization and (2) a standalone production rescheduling comparison by ID-price scenario.

The offering model's DA + ID + up-reserve case has expected modeled net cost €129,533.65, compared with €138,744.23 for DA-only plus RT imbalance settlement. That is a modeled market-participation difference of €9,210.58 (6.6%).

Separately, the production rescheduler estimates probability-weighted cost reduction of €2,334.04 versus fixing the Stage 1 jobs, under scenario-matched ID prices and the same MT commitment. This is about 1.70% of the standalone fixed-plan comparison cost.

Do not add these two savings figures. The rescheduling result is a separate counterfactual, not a benefit already optimized inside the offering model. The combined report is an analysis of both, not a combined integrated-solver result.

All amounts are modeled energy-market/operating costs, not factory profit or ROI: product-sale revenue and other non-energy business costs are excluded.

Indicator	Saved run result
Market case cost (DA + ID + up-reserve)	€129,533.65
DA-only market-case cost	€138,744.23
Market-case difference	€9,210.58
Separate weighted reschedule difference	€2,334.04
BESS degradation rate / expected cost	€5/MWh throughput / €747.45

How to interpret the two savings results

The offering optimization chooses production, DA bids/offers, ID positions, reserve offers, dispatch, and settlements within its model. It reports expected market/operating cost under the modeled scenario tree.

The separate rescheduler takes the Stage 1 job schedule and MT commitment, then re-solves production and operating dispatch for each ID scenario. It compares the resulting cost with a fixed-job-plan solve using that same scenario's ID prices and commitment. The rescheduler does not feed those schedules back into DA offers, the offering objective, or RT offering results.

The market comparison moves from DA-only to DA+ID by €1,031.88, then from DA+ID to DA+ID+reserve by €8,178.70. The complete market-ladder difference is €9,210.58. The separate reschedule comparison is €2,334.04. They use different counterfactuals and scopes, so their arithmetic sum is not a validated combined saving.

An integrated value would require a single re-optimization in which scenario-contingent production schedules affect market positions and all settlements, with consistent output/service constraints. No such integrated result is claimed in this report.

Comparison	Baseline	Alternative	Difference
Market ladder: DA-only to fu	€138,744.23	€129,533.65	€9,210.58
Standalone ID rescheduler:	€137,259.68	€134,925.65	€2,334.04

Market offering results

The same 3 ID-price × 3 RT scenario tree is used in the three market-stage cases; RT imbalance settlement remains active in every case. Adding ID trading reduces the baseline modeled cost modestly. Adding the modeled up-reserve option accounts for the larger incremental reduction in this run.

The full case records €16,602.20 combined reserve capacity and activation revenue, while its incremental cost reduction versus DA+ID is smaller. Gross reserve revenue is not equal to net value: the optimization also changes energy positions and operating decisions.

Full-case expected cost drivers are DA net settlement €83,067.40, ID net settlement €29,439.00, reserve capacity revenue −€12,071.82, reserve activation revenue −€4,530.38, imbalance €3,860.31, MT cost €28,408.53, BESS degradation €747.45, and demand-response cost €613.16.

The model terminated optimally with a reported MIP gap of 0.986% and independent verification passed. The scenario set is small, so these results are scenario-study outputs, not a robust statistical forecast.

Market participation case	Expected net cost	Saving vs. DA-only	MIP gap
DA only + RT imbalance set	€138,744.23	€0.00	0.852%
DA + ID + RT imbalance set	€137,712.35	€1,031.88	0.946%
DA + ID + RT settlement +	€129,533.65	€9,210.58	0.986%

Standalone ID production rescheduling results

All three ID scenarios have probability 1/3. Scenario 0 reduces the matched fixed-plan cost by €6,069.81; scenario 1 reduces it by €932.30; scenario 2 retains the fixed plan and has zero saving. Their probability-weighted result is €2,334.04.

The scenario-0 and scenario-1 schedules each change 13 jobs. Scenario 2 changes none because the fixed Stage 1 plan is retained as the least-cost incumbent. All reported reschedule solves passed their recorded verification.

Important output caveat: Stage 1 and scenarios 0 and 2 produce 193 units, but scenario 1 produces 192. Its €932.30 lower cost therefore is not a strict same-output comparison; the one-unit output difference may contribute to the cost reduction. Treat the weighted €2,334 as a scheduling-study estimate, not a pure timing-only saving. Re-run with equal production requirements before attributing all cost difference to flexibility.

At cutoff hour 0, no jobs have started and none are frozen. At a later cutoff, already-started jobs are fixed. This run's reschedule MIP gaps are about 0.97%, 0.93%, and 0% for scenarios 0, 1, and 2.

ID scenario	Probability	Fixed-plan cost	Rescheduled cos	Cost saved	Jobs changed	Units
0	33.3%	€127,602.00	€121,532.19	€6,069.81	13	193
1	33.3%	€146,511.78	€145,579.48	€932.30	13	192
2	33.3%	€137,665.27	€137,665.27	€0.00	0	193
Weighted / expe	100.0%	€137,259.68	€134,925.65	€2,334.04	—	192.7 (expected)

BESS, risk, and decision-use caveats

BESS degradation was explicitly set to €5/MWh on a throughput basis. The model charges throughput for scheduled charging after activation curtailment and for scheduled plus reserve-activated discharge. The saved expected degradation cost is €747.45; this is not the full cost of buying energy to charge the battery.

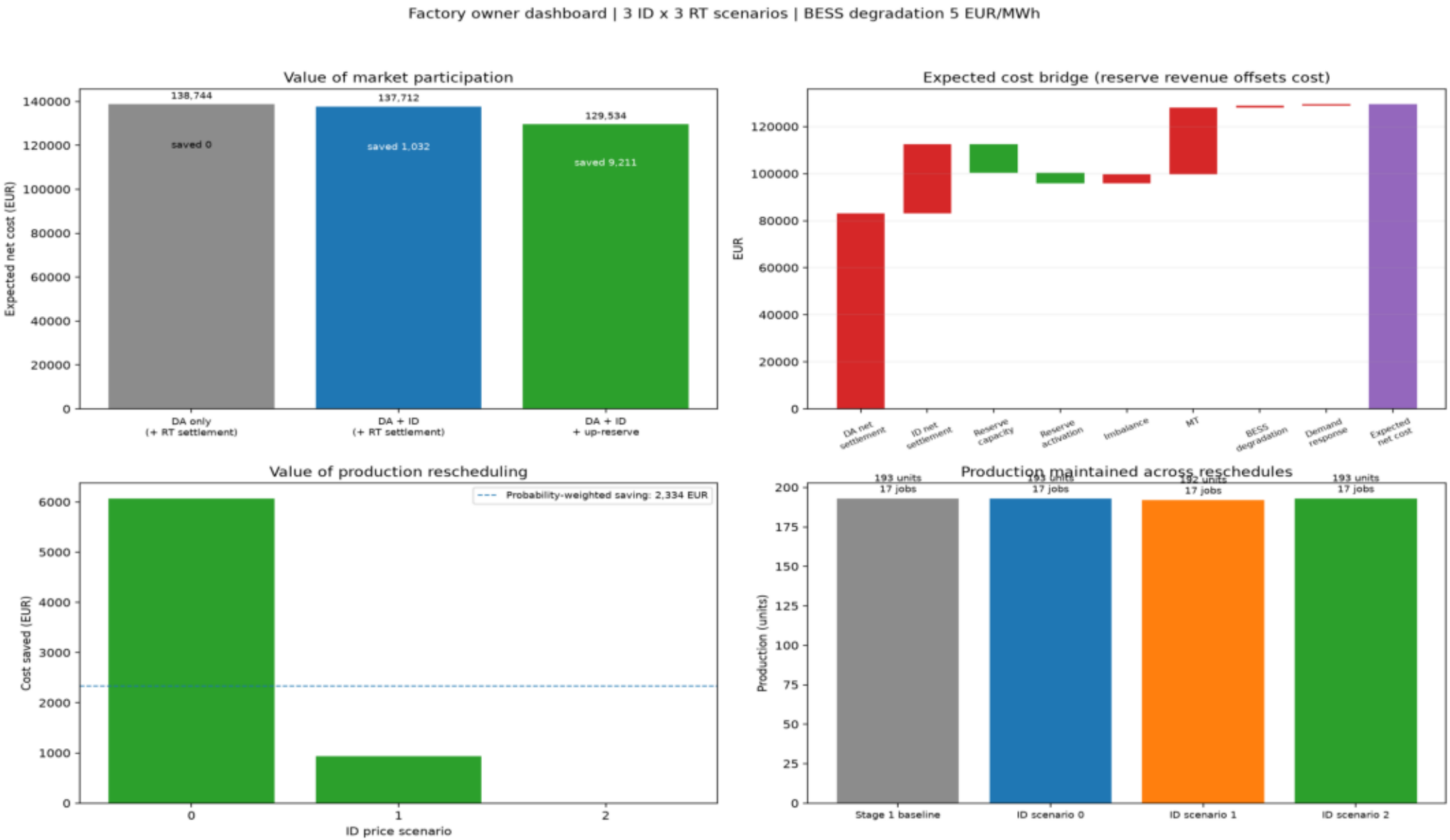
The run contains nine joint paths. beta=0 means CVaR is reported but does not influence this run's objective; with only nine paths, the 95% tail estimate is especially sample-sensitive.

The modeled reserve is generic upward reserve, not proof of a qualified market product or a compliant Danish bid. Validate current market rules, technical qualification, gate closure, and settlement details before operational use.

For a stronger rescheduling business case: enforce equal required output and service levels in fixed and recourse cases; use more realistic ID execution constraints; increase scenario coverage; then integrate production recourse with the offering and settlement model before reporting a combined value.

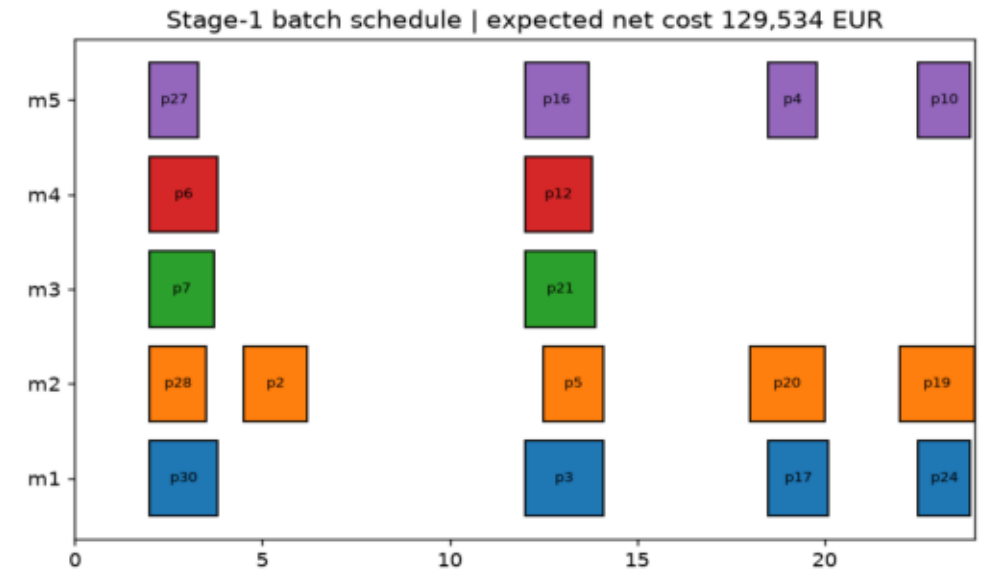
Run setting	Value
BESS degradation	€5/MWh, throughput basis
Expected BESS degradation cost	€747.45
Expected reserve offer / active hours	36.15 MW / 23 hours
Expected activated reserve	70.99 MWh
Scenario tree / CVaR weight	3 ID × 3 RT = 9 joint paths / beta = 0
Offering verification	Passed

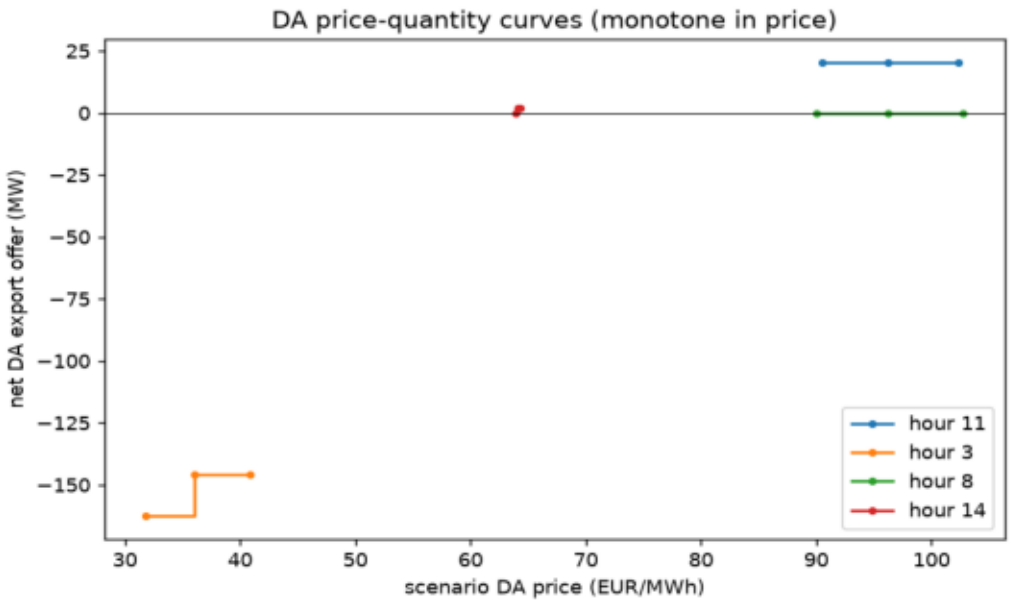
Owner dashboard

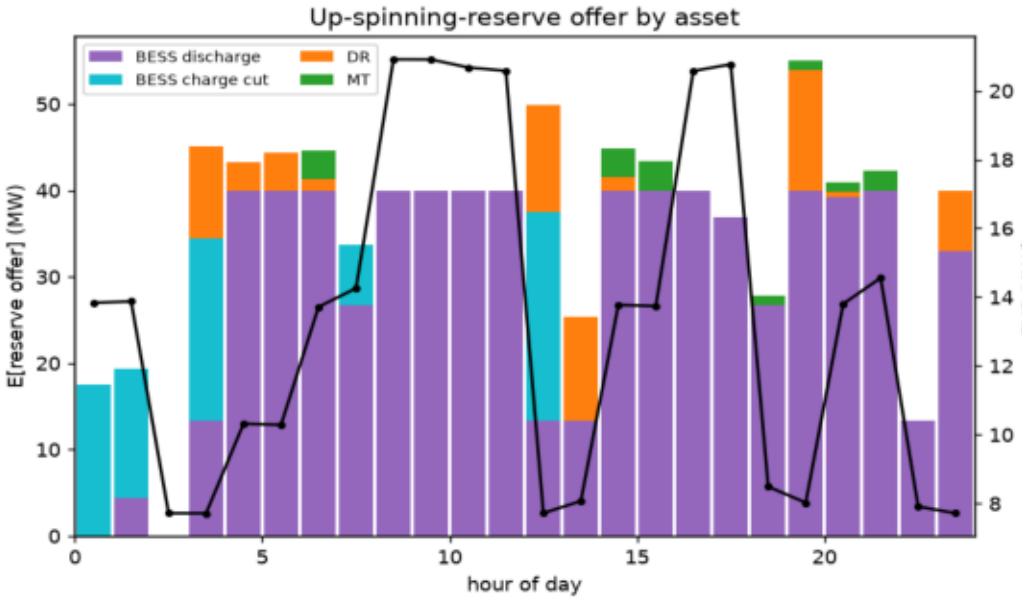


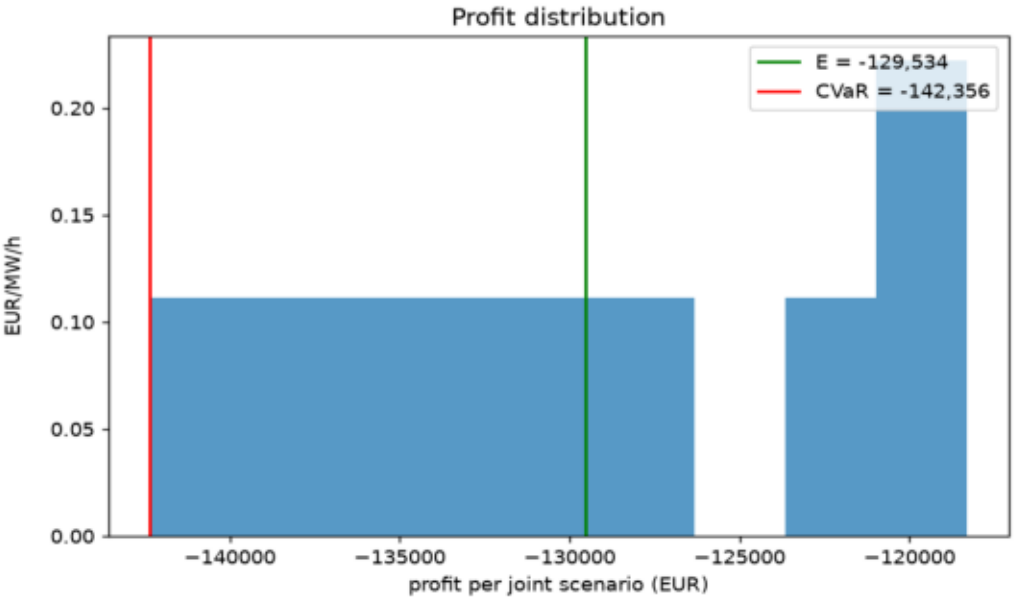
Four-part overview of market participation costs, the operating-cost bridge, standalone rescheduling value, and production volumes.

Offering overview



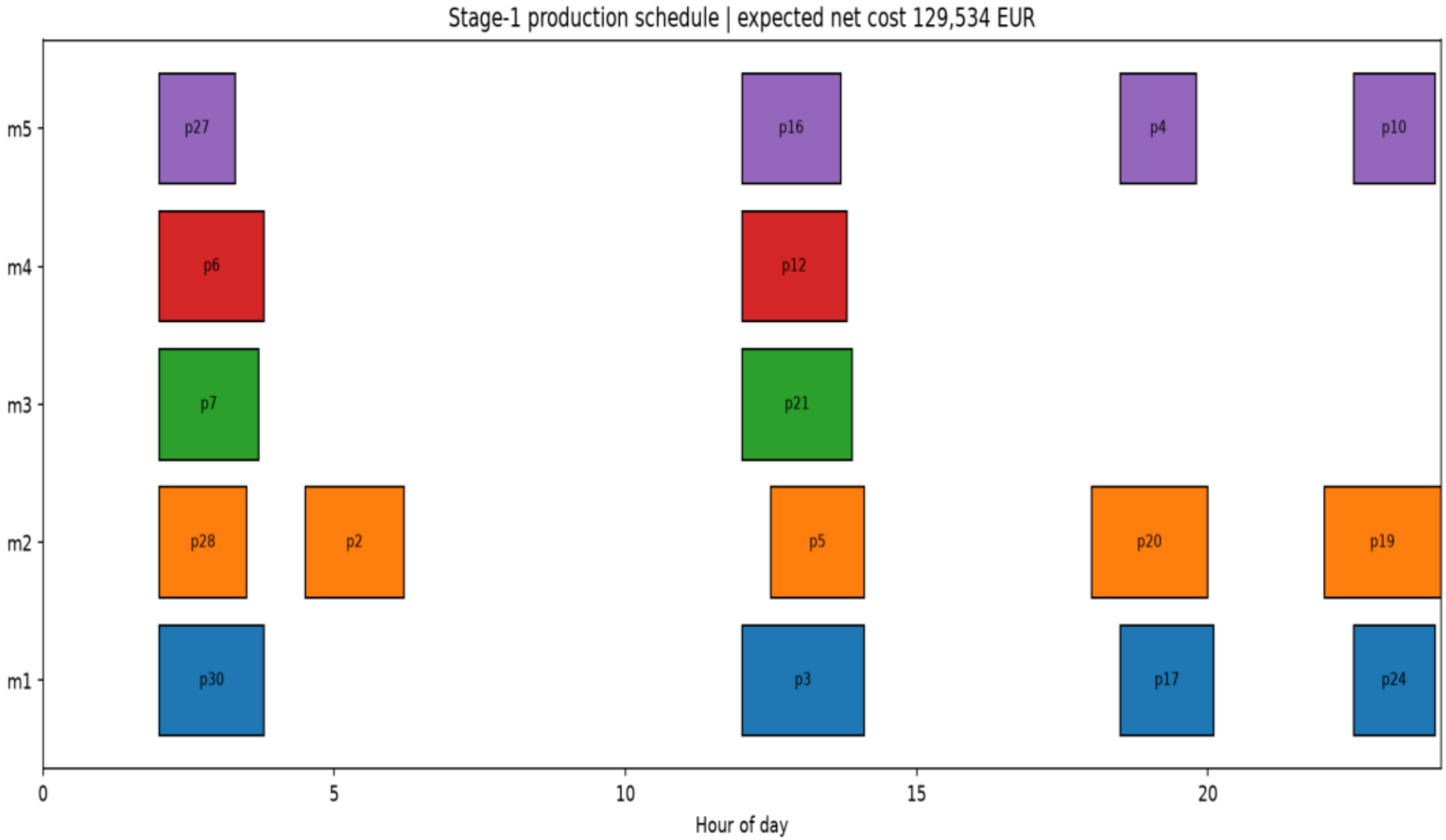






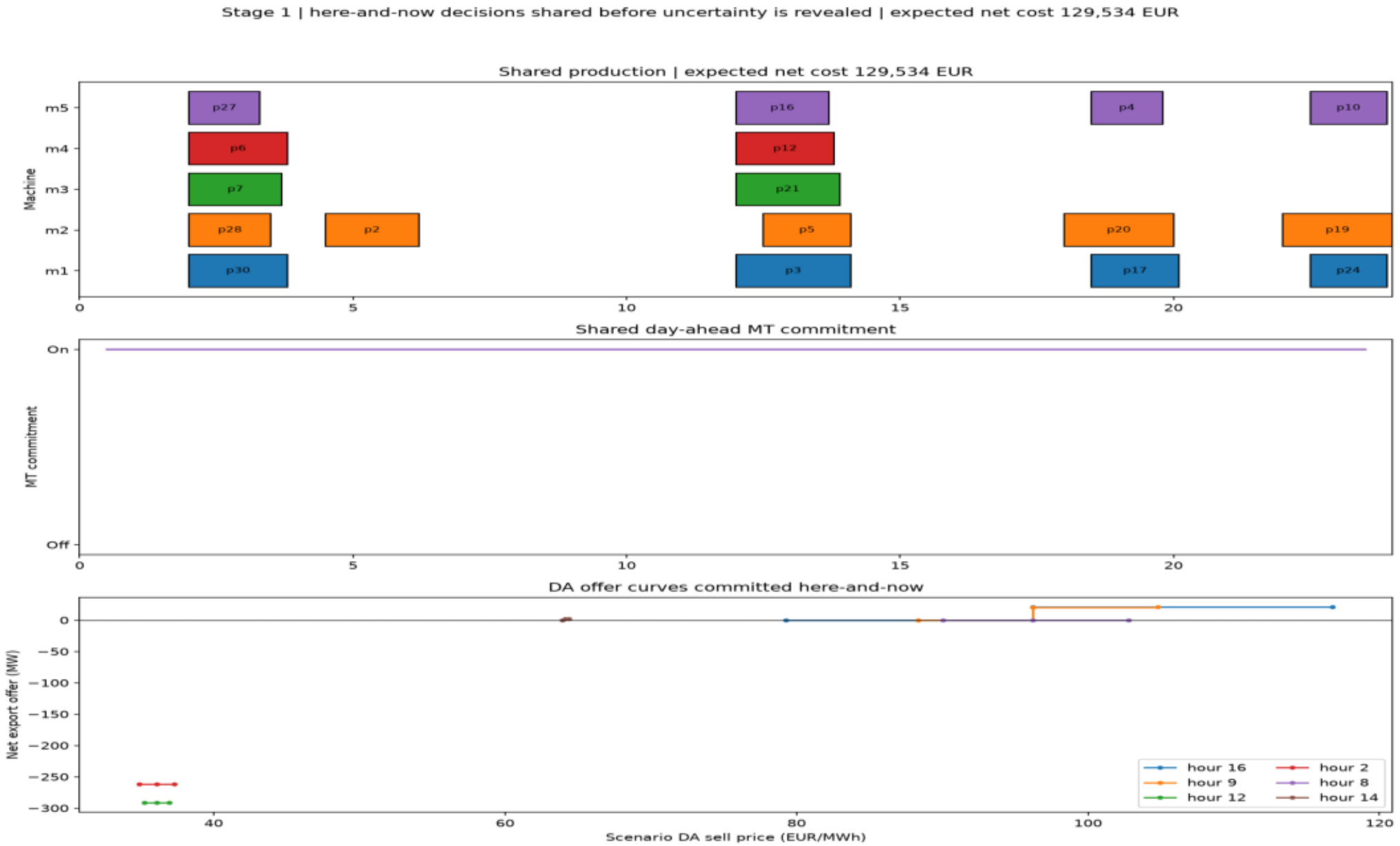
Compact view of the Stage 1 production plan, DA price-quantity curves, reserve offers by asset, and scenario profit distribution.

Stage 1 production schedule



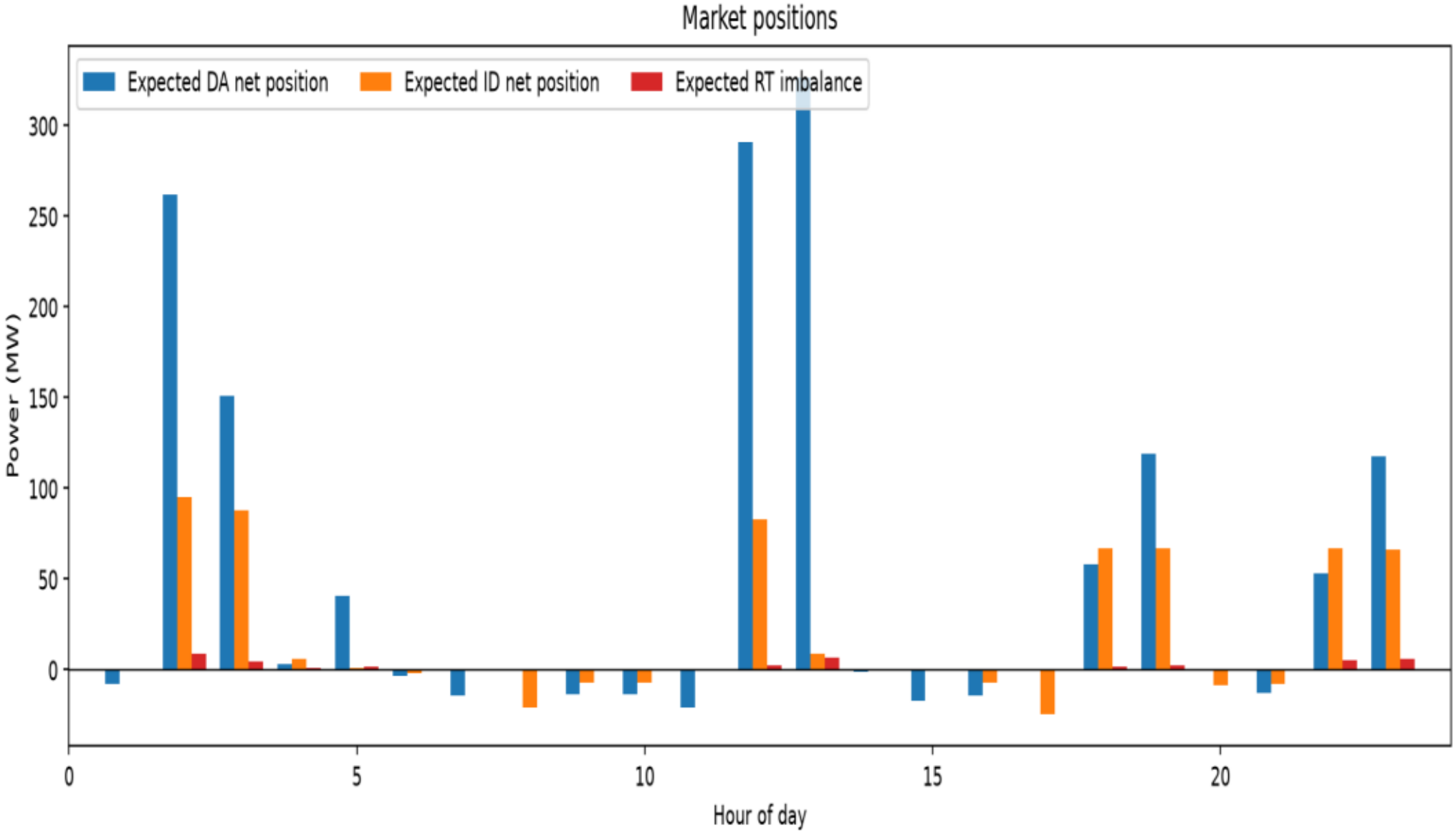
Shared production schedule before uncertainty is revealed. Bars show jobs by machine, start time, and duration.

Stage 1: here-and-now decisions



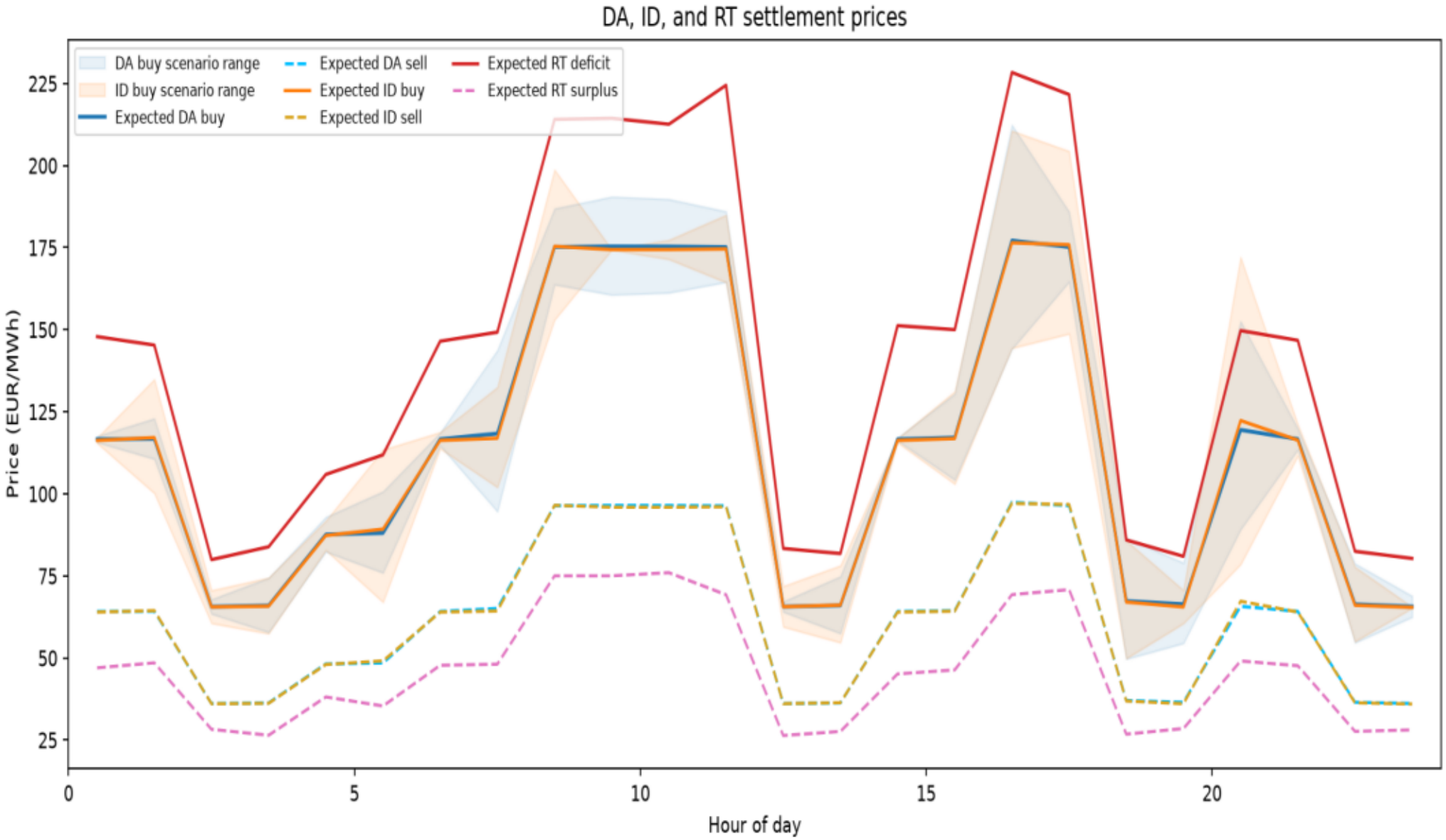
Shared production, day-ahead microturbine commitment, and selected DA offers.

Expected market positions



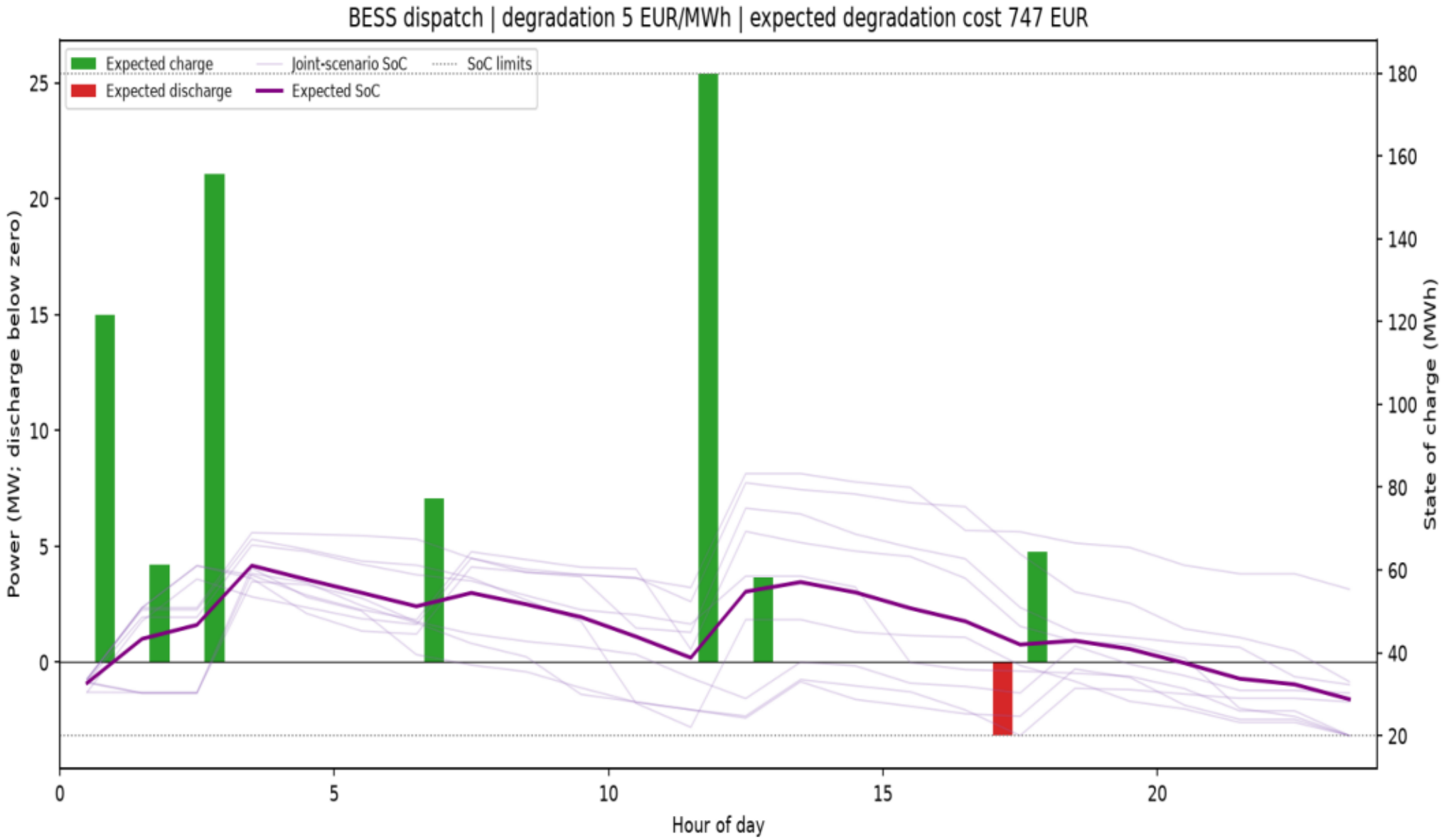
Expected net DA, ID, and real-time positions. Positive values indicate net import/deficit; negative values indicate net export/surplus.

Market settlement prices



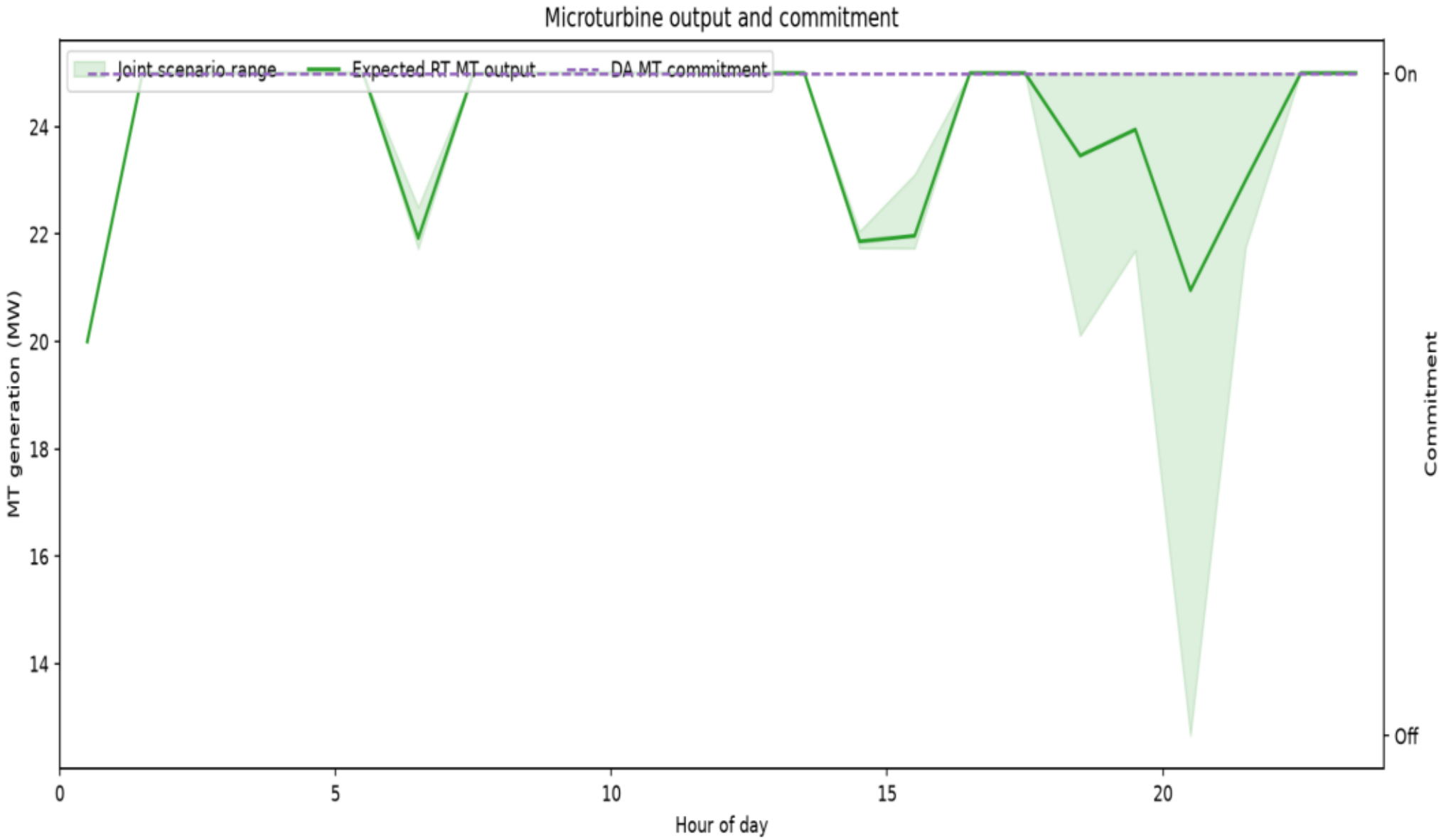
Expected DA, ID, and RT prices by hour, with scenario ranges for DA-buy and ID-buy prices.

BESS dispatch and state of charge



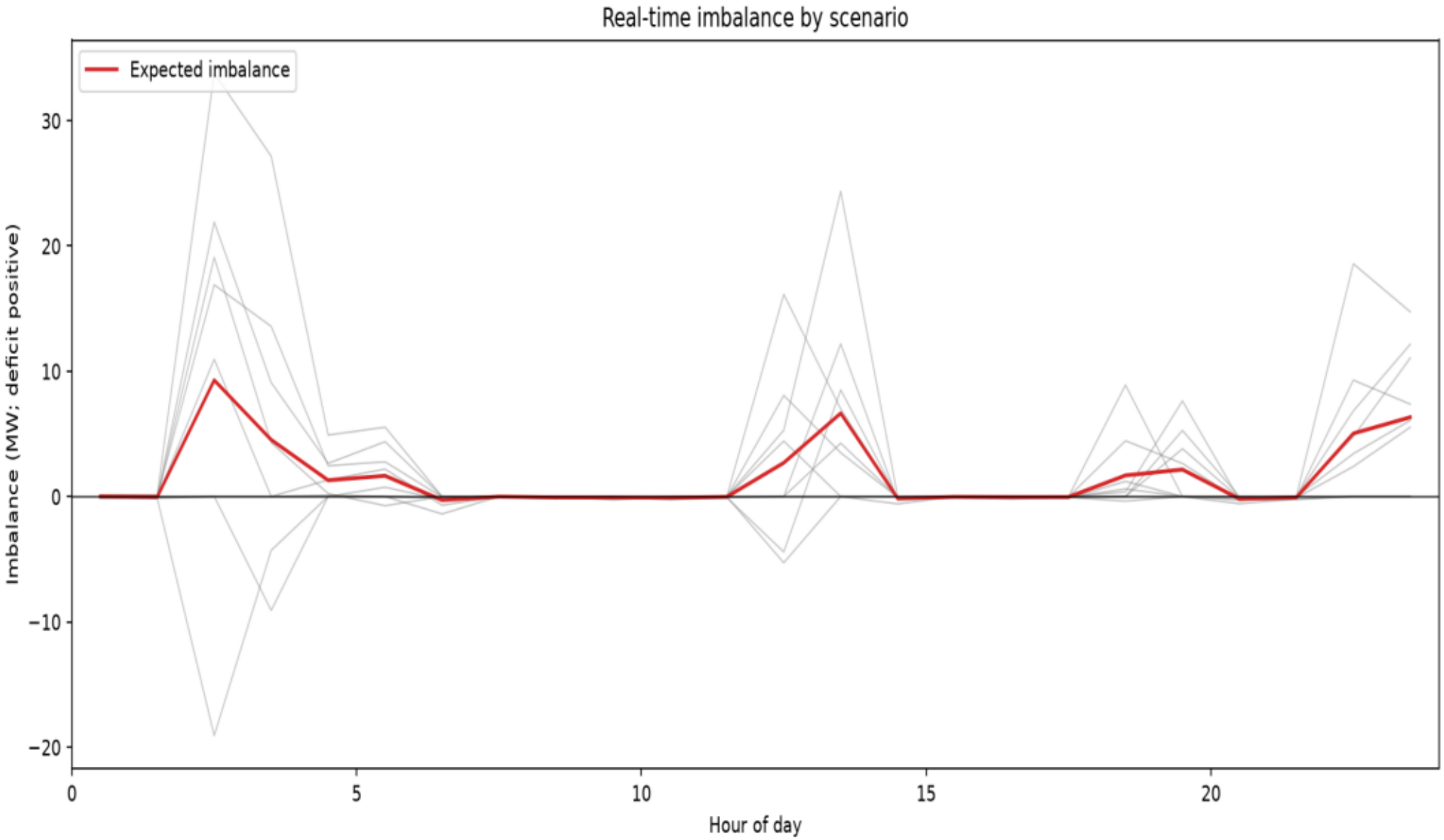
Expected charge/discharge and joint-scenario state of charge. The run charges €5/MWh on throughput and reports €747.45 expected degradation cost.

Microturbine dispatch



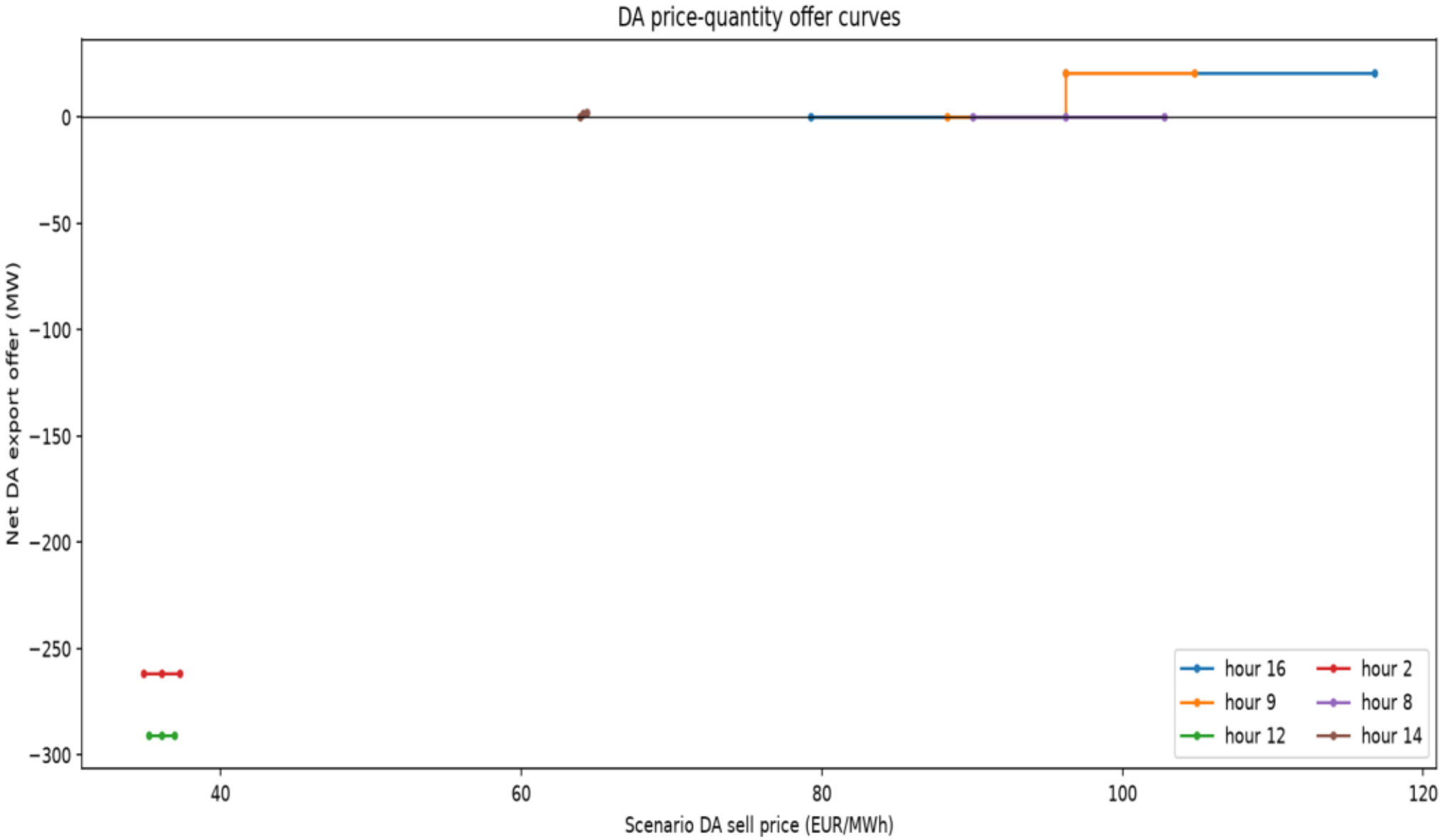
Expected RT output and scenario range, alongside the shared DA on/off commitment.

Real-time imbalance



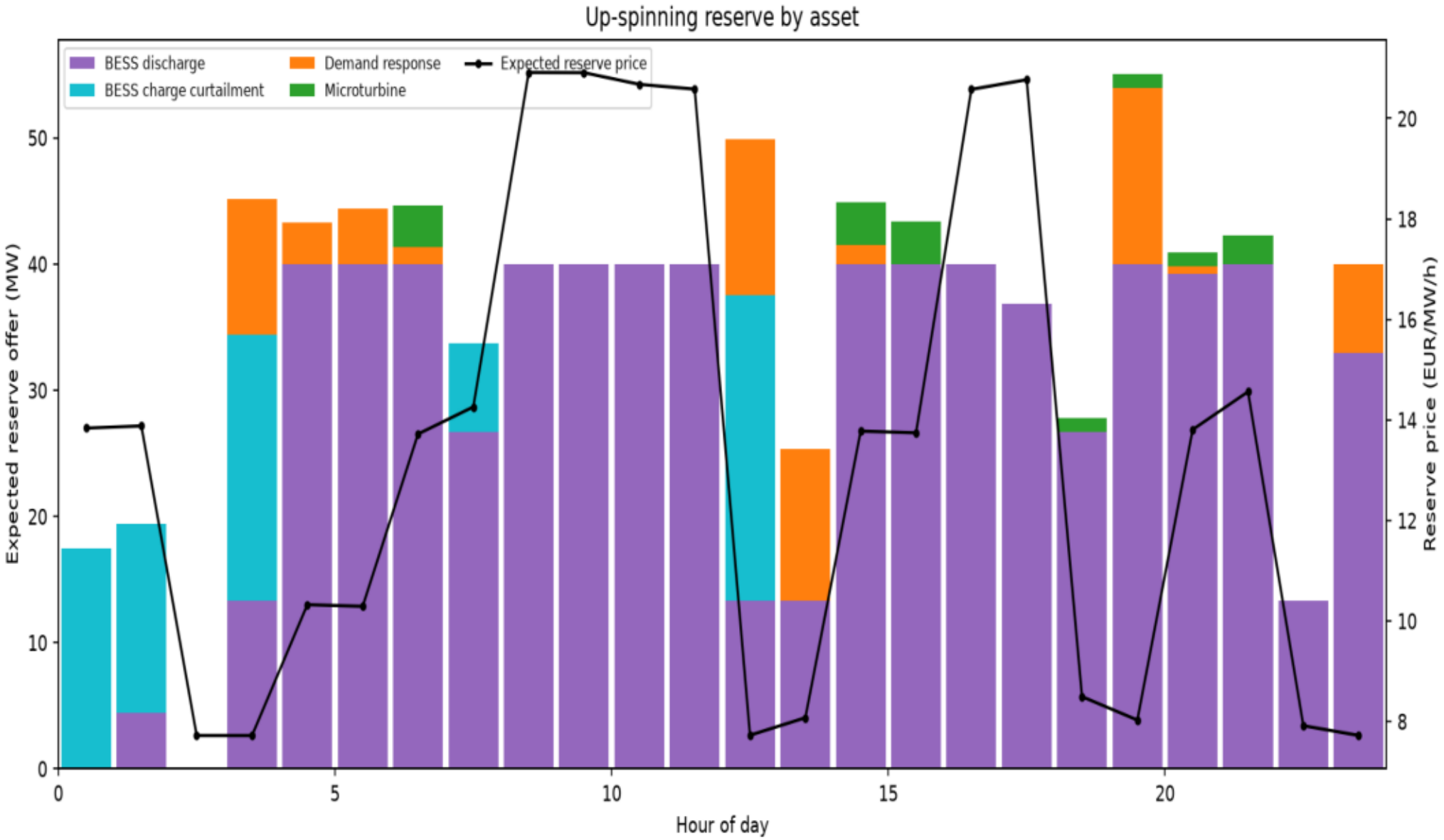
Scenario and expected deficit-minus-surplus. Values below zero represent surplus, not a negative physical quantity.

Day-ahead offer curves



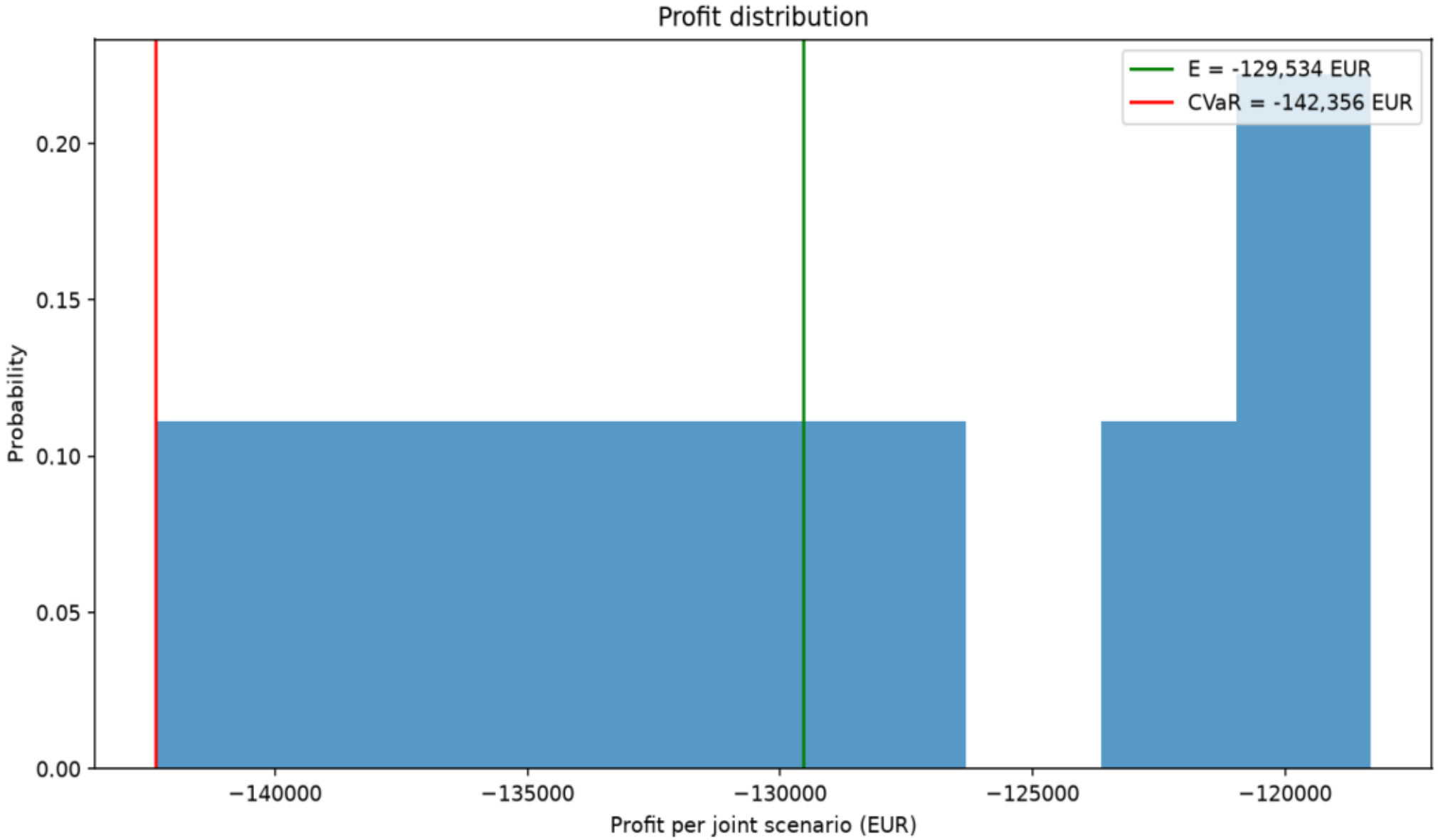
Net DA export is sell quantity minus buy quantity. Negative values therefore mean net import; individual buy/sell quantities remain nonnegative.

Upward-reserve offers



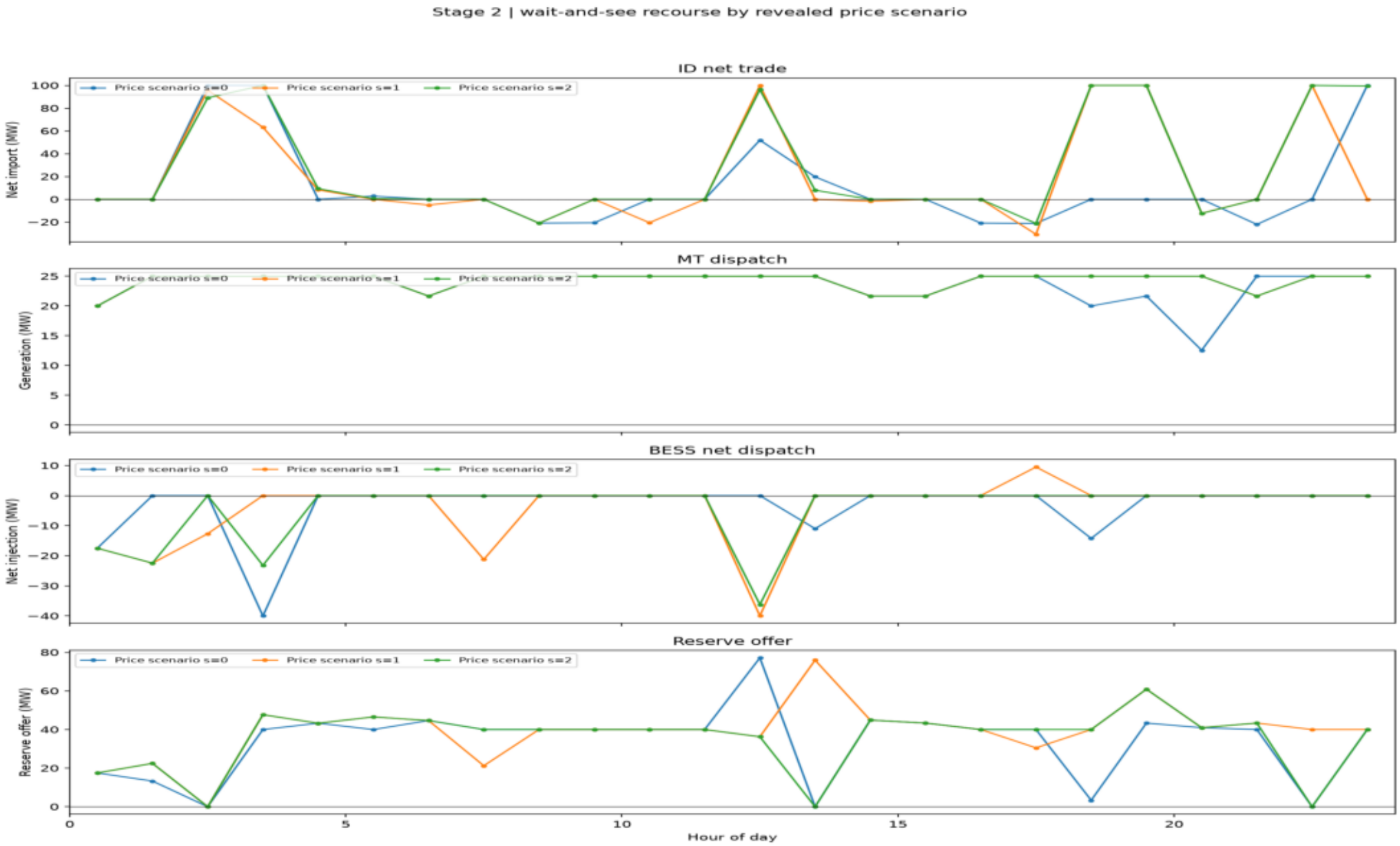
Expected reserve capacity by asset, with expected reserve price on a second axis. Offers are capacity and need not equal activated energy.

Scenario profit distribution



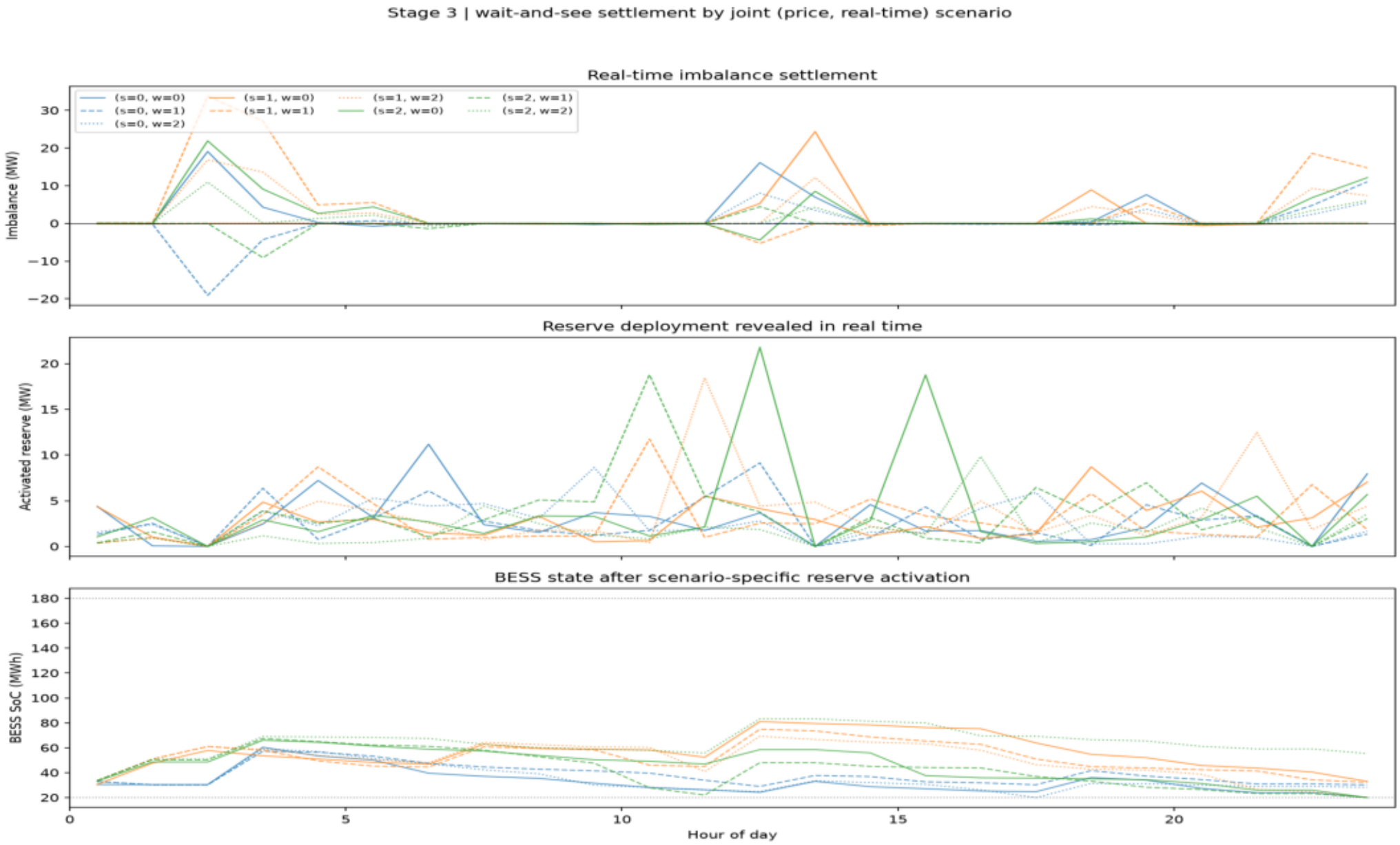
Probability-weighted modeled profit with expected-profit and CVaR markers. The objective excludes product-sales revenue and other non-energy business costs.

Stage 2: price-scenario recourse



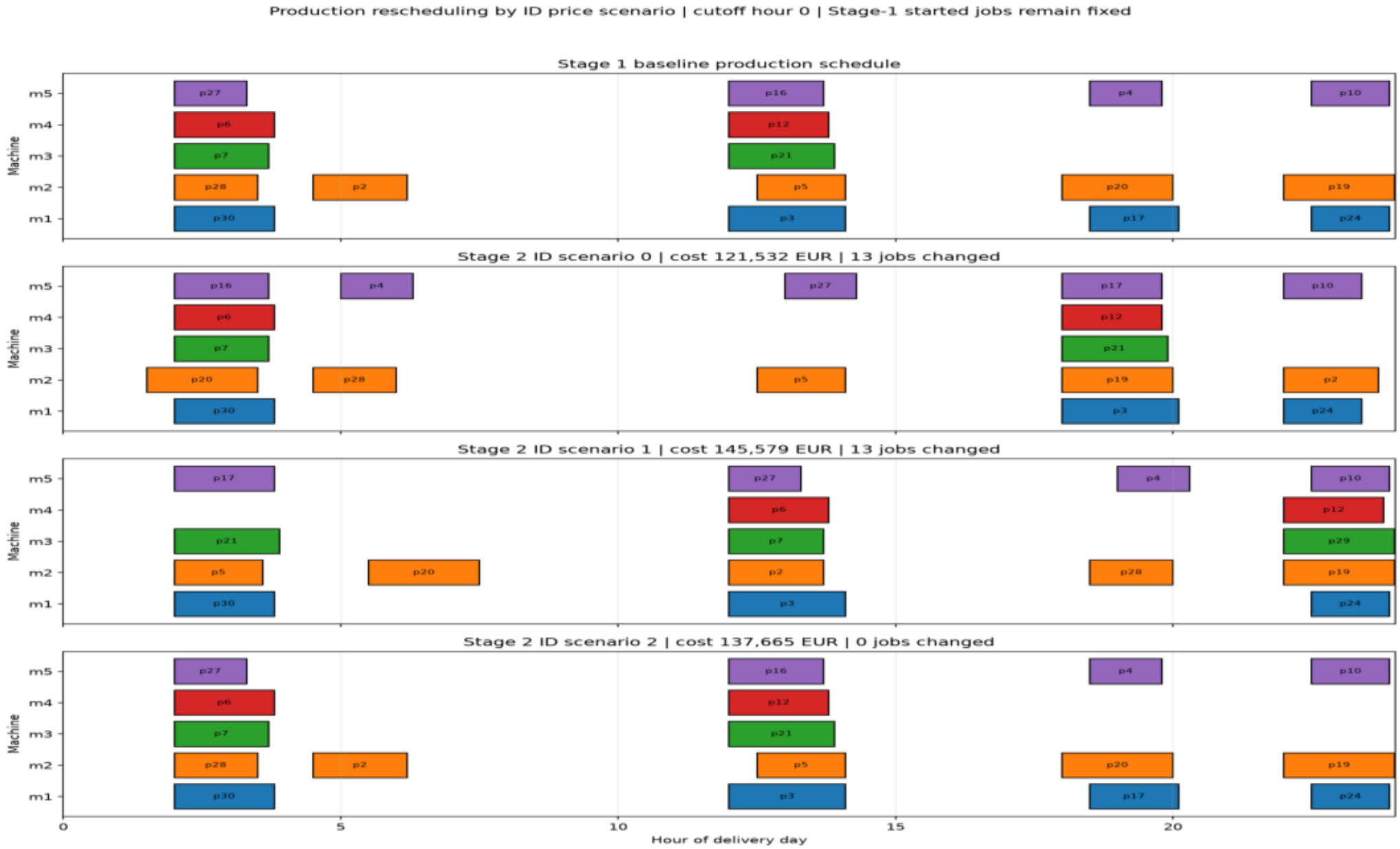
Intraday net trade, MT dispatch, BESS net dispatch, and reserve offer for each price scenario after that scenario is revealed.

Stage 3: real-time recourse



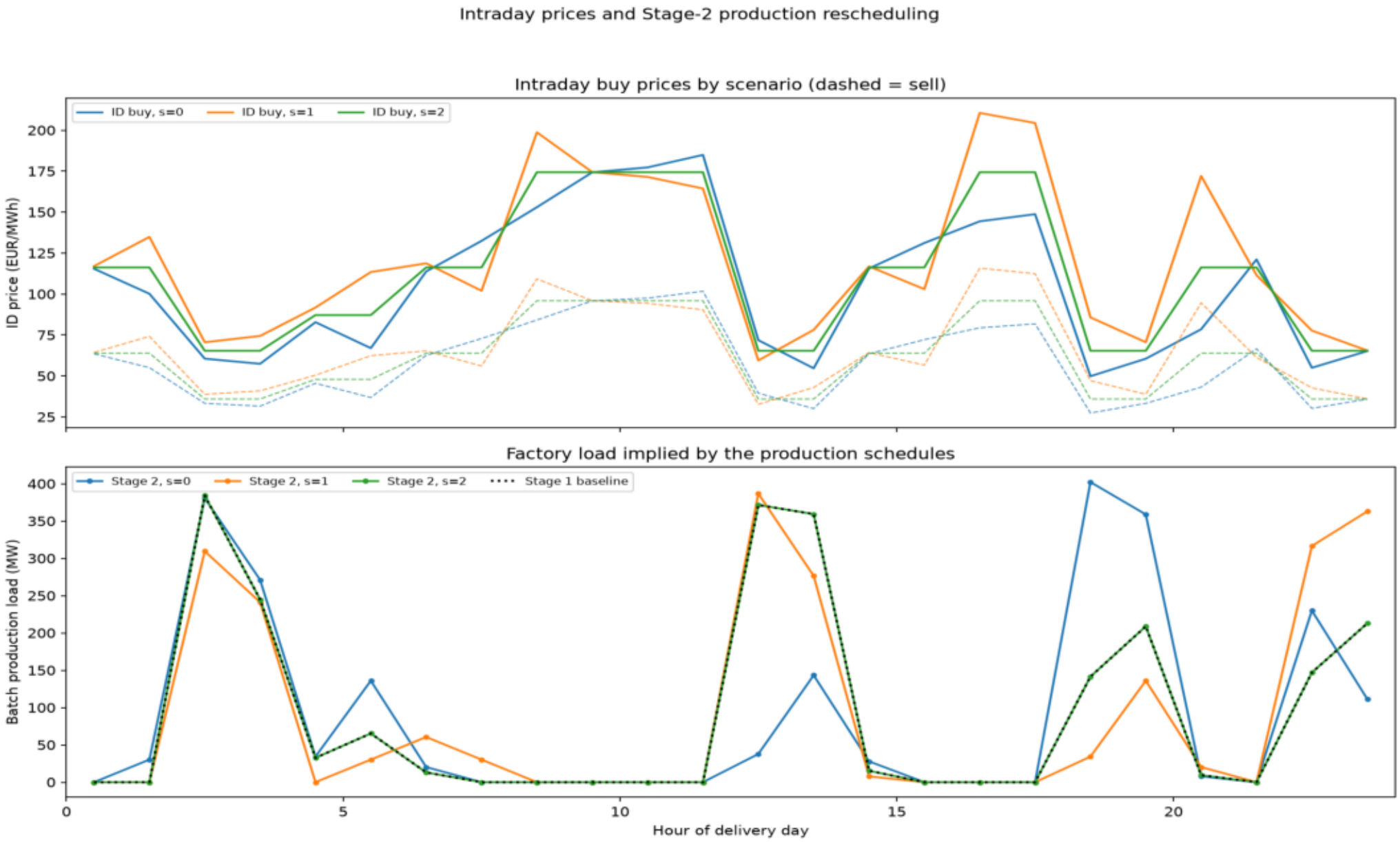
Joint-path imbalance, reserve activation, and BESS state of charge after real-time uncertainty is revealed.

Standalone production reschedules



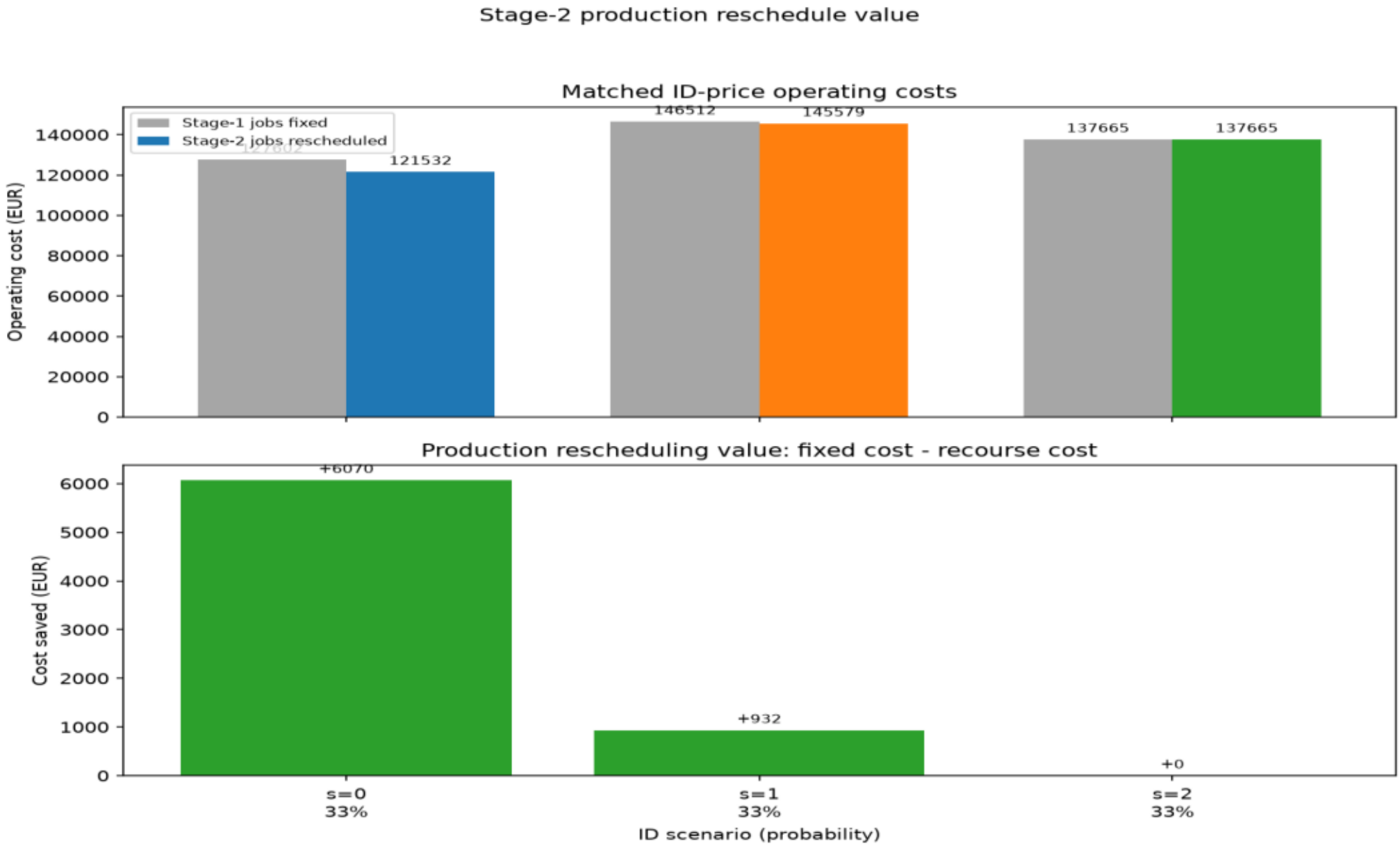
Stage 1 schedule followed by a separate Stage 2 production plan for each ID scenario. This is not integrated into the offering-profit optimization.

ID prices and implied production load



Intraday price scenarios and resulting Stage 2 production load, compared with the Stage 1 baseline.

Matched rescheduling cost comparison



Scenario-by-scenario fixed-plan and rescheduled costs and their difference. Scenario 1 produces one fewer unit, so its cost reduction is not a strict same-output comparison.